

The Log4j Issue: Do You Know What is in Your Software?

In December 2021, a vulnerability with a severity score of 10 out of 10 in a widely used logging library affected the security of digital systems across the Internet. The vulnerability was in the Log4j logging tool. This article explores the importance of knowing and documenting the usage of different components in your software.



Log4j is an open source Apache logging framework that developers use to keep a record of the activity within an application. The open source software creates a built-in log or record of the activity that software developers can use to troubleshoot problems or track data within their programs.

You can think of this software as a diary. Its job is to keep records of how online services and applications are being used.

The vulnerability within Log4j has been coined Log4Shell. It enables a remote attacker to take control of a device on the Internet if the device is running certain versions of Log4j 2.

A former cybersecurity commissioner for the Obama Administration says hackers have the capacity to, “Spy on the activities of the users of [affected] systems. They have the capacity to use that system to island-hop into other systems. They have the capacity to become disruptive. It really varies.”

Within four days of the vulnerability being discovered, 840,000 attacks were launched and some hackers had already developed tools that automatically attempted to exploit the bug. During the first 72 hours after this vulnerability went public, reports documented up to 100 attempted attacks per minute.

A high-profile attack deriving from Log4j was launched against one of the largest Vietnamese crypto trading platforms Onus. Hackers successfully planted backdoors to retrieve sensitive databases before Onus patched its systems. They demanded a US\$ 5 million ransom, which Onus refused to pay, so they put the data of nearly 2 million Onus customers up for sale on different forums.

Reports from companies such as Google and Microsoft have indicated that government hacking groups around the world are leveraging the Log4j vulnerability for attacks. According to Microsoft, state-sponsored hackers from China, Turkey,